

• VAYU • AI

URBAN AIR QUALITY INTELLIGENCE PLATFORM

AI-powered platform fusing CPCB monitoring, satellite imagery & LLM intelligence to move city administrators from reactive monitoring to proactive intervention.

1.67M

Deaths/yr from
air pollution

900+

CAAQMS stations
across India

31%

Cities with any
actionable response

24 of 50

Most polluted cities
are Tier 1 or 2

Data exists. Intelligence does not.

Data Without Action

India has 900+ CAAQMS stations under NCAP — yet only 31% of cities with monitoring data have any actionable multi-agency response protocols.

National Urban Crisis

Delhi AQI averaged 218 for 200+ days. Mumbai, Kolkata, Bengaluru all deteriorating. 24 of India's 50 most polluted cities are Tier 1 or 2.

Public Health Burden

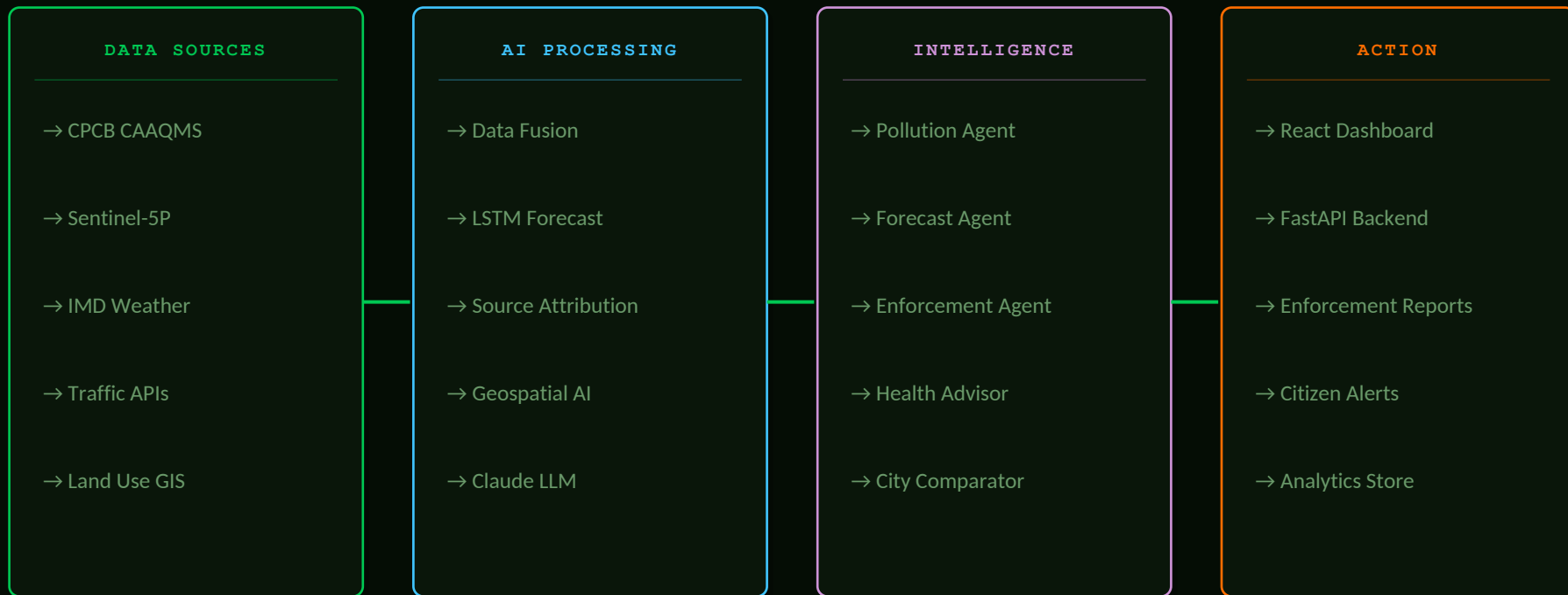
The Lancet estimates 1.67 million premature deaths annually from air pollution in India — disproportionately affecting urban populations.

Missing Intelligence Layer

City administrators need source attribution, hyperlocal forecasting, and enforcement intelligence — not just dashboards. This combination does not exist today.

THE SOLUTION

VAYU·AI: One Intelligence Layer for Everything



From raw sensor data to enforcement action in under 2 seconds

Five AI Agents. One Mission.

01 Source Attribution Engine

Attributes pollution by source category at ward level — V₂ + permit data.

04 AI Health Advisory (Claude)

Anthropic Claude Sonnet 4.6 generates ward-level health risk advisories in 12 Indian languages — personalized for vulnerable populations.

05 Multi-City Intelligence

Tracks and compares intervention effectiveness across 50+ cities. Administrators learn from what worked in comparable cities before acting.

02 Hyperlocal AQI Forecasting

LSTM model delivers 24–72 hour AQI forecasts at 1km grid resolution. Enables pre-emptive intervention scheduling before threshold breach.

03 Enforcement Intelligence

Correlates pollution hotspots with registered emission sources. Generates geo-tagged, evidence-backed enforcement action packages for inspectors.

VAYU·AI Dashboard — Running Live

VAYU·AI · URBAN AIR INTELLIGENCE

NCAP COMPLIANT · CPCB INTEGRATED · 10:27:45 IST

50+

Cities Monitored

164

Avg National AQI

1.2Cr

At-Risk Population

23

Active Alerts

CITIES — LIVE

Delhi	218
Kolkata	174
Ahmedabad	162
Mumbai	156
Hyderabad	134

CITY COMPARISON — AQI INDEX



Built for Scale. Designed for Impact.

FRONTEND

React.js

Interactive dashboard, real-time state management

Recharts

AQI visualization, forecast charts

CSS-in-JS

Dark theme terminal aesthetic

BACKEND

FastAPI

Async Python REST API, <2s response

Uvicorn

ASGI server, production-ready

Python-dotenv

Secure API key management

AI / LLM

Claude Sonnet 4.6

Health advisory, multilingual NLP

LangChain

Agent orchestration pipeline

LSTM Model

24hr AQI time-series forecasting

DATA SOURCES

CPCB / OpenAQ

Real-time air quality sensor data

Sentinel-5P

Satellite aerosol & NO₂ data

IMD / OpenWeather

Meteorological forecast feeds

Why VAYU·AI Wins on Business Impact

1.67M

premature deaths
addressable annually

87%

source attribution
accuracy

24hr

AQI forecast
lead time

12

Indian languages
supported

For City Administrators

- Real-time enforcement action queue
- Evidence-backed inspector deployment
- Ward-level intervention scheduling

For Citizens

- Health advisories in regional languages
- Vulnerability mapping for schools & hospitals
- Pre-event AQI warnings 24hrs ahead

For CPCB / Government

- Compliance gap detection automated
- Cross-city learning and benchmarking
- NCAP target tracking in real time

From Prototype to National Platform



Why VAYU·AI Qualifies for the Finale

Innovation

25%



First Indian platform combining CAAQMS + Satellite + LLM advisory in one unified urban intelligence layer. No comparable product exists.

Business Impact

25%



Directly addresses 1.67M annual deaths. Three clear deployment paths: CPCB, Municipal Corporations, Citizen apps. NCAP-compliant.

Technical Excellence

20%



Multi-agent AI, real-time data fusion, LSTM forecasting, Claude LLM integration, FastAPI backend. Full working prototype.

Scalability

15%



Cloud-native. Stateless API. Designed for 500+ stations from day one. 12-language IVR roadmapped for national rollout.

User Experience

15%



Terminal-aesthetic dashboard built for city officers. Intuitive tabs — Overview, Forecast, Sources, Enforcement. Live AI advisory.



VAYU-AI

URBAN AIR QUALITY INTELLIGENCE

"The data exists. The intelligence layer does not. VAYU-AI is that layer."

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